

CSCI218: Foundations of Artificial Intelligence



UNIVERSITY
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AUSTRALIA

Flowers Recognition (KNN, **MLP** and CNN)

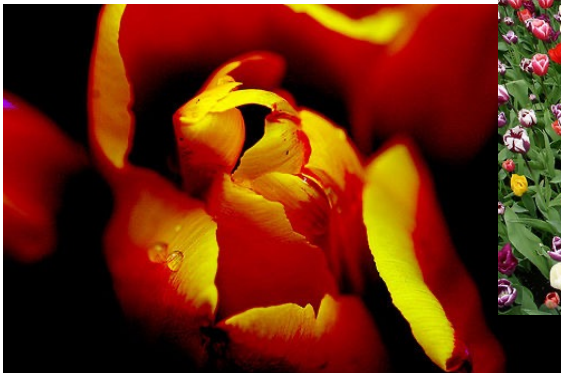


Outline

- Background
- Dataset
- Task
- Approach

Background

- Several available apps which can identify flowers and plants, such as Microsoft's Bing and Google plant identifier, Google lens.
- Automation of flower classification can boost flower searching for public and floriculture, especially in special scenario, like digital library.
- Large intra-class variation and small inter-class variation among different classes of flower makes this a challenging task.



Dataset – flower images

- This dataset contains 4317 images of flowers, based on data flicr, google images, Yandex images.
- Here are 5 classes: daisy, tulip, rose, sunflower, dandelion
- About 800~1000+ photos for each class with various pixels resolution and proportions.



Machine Learning -- KNN

- Import modules
- Extract Image features -- [Color Histogram](#) approach
- Divide dataset into training set, validation set, and test set
- Building a classifier with KNN algorithm
- Evaluation on test set by performance metrics

Modules

- Data visualization
 - Matplotlib
 - Seaborn
- Data processing
 - CV2 (opencv-python)
 - Numpy (Python library for working with arrays)
- Machine Learning libraries
 - Scikit-Learn
- Model performance metrics
 - Scikit-Learn

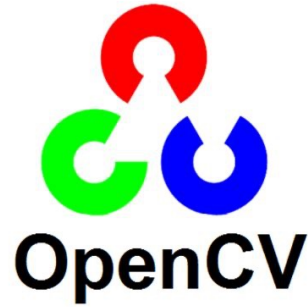
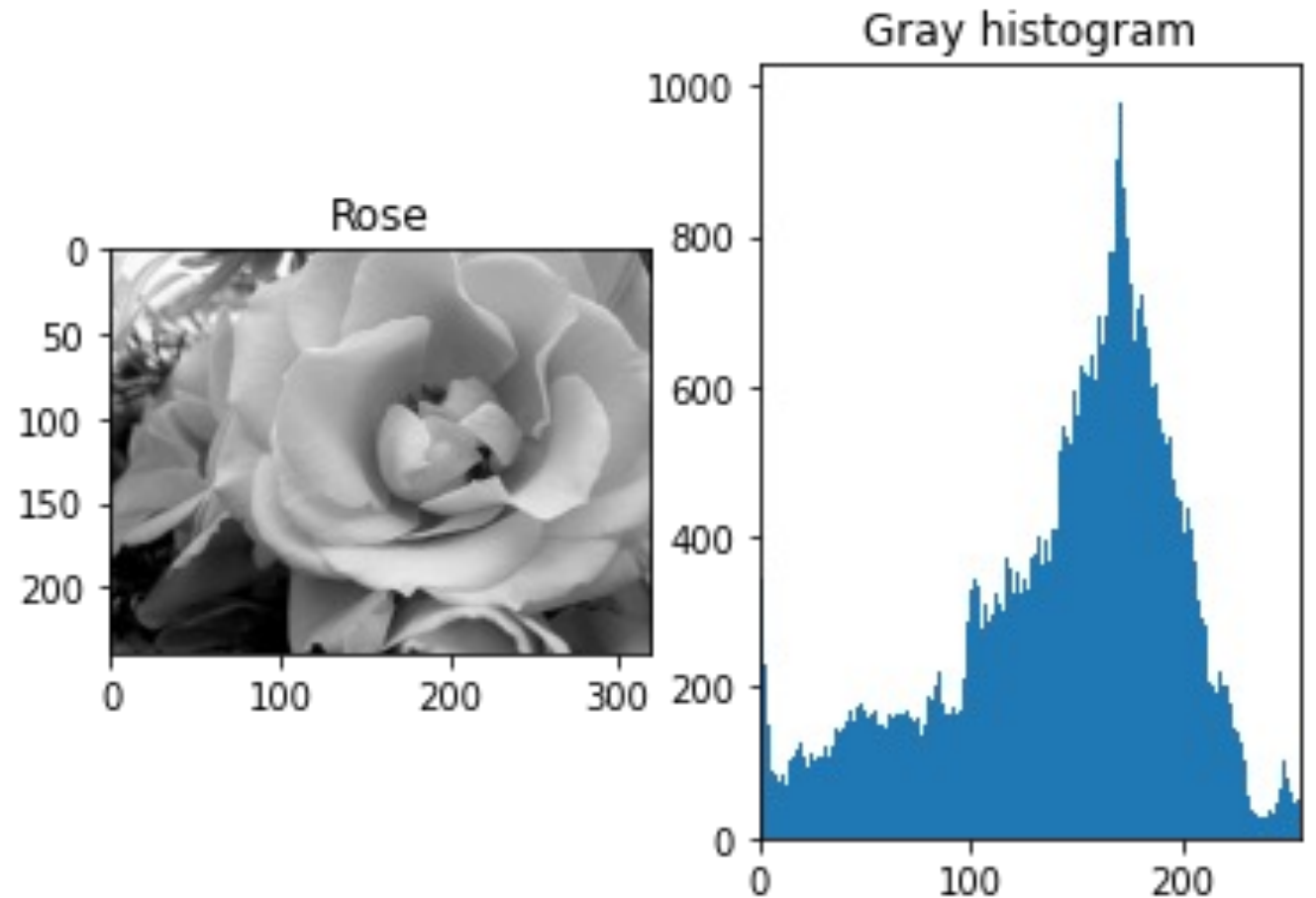


Image feature

- Color Histogram

A plot with pixel values (ranging from 0 to 255, not always) in X-axis and corresponding number of pixels in the image on Y-axis.



Left region of histogram shows the number of darker pixels in image and right region shows the number of brighter pixels.

This histogram is drawn for grayscale image.

Image feature

- RGB Histogram
 - an acronym for "Red Green Blue"
As a colour model representing color.
- A digital image has a red, green and blue channel.
- A grayscale image has just one channel.

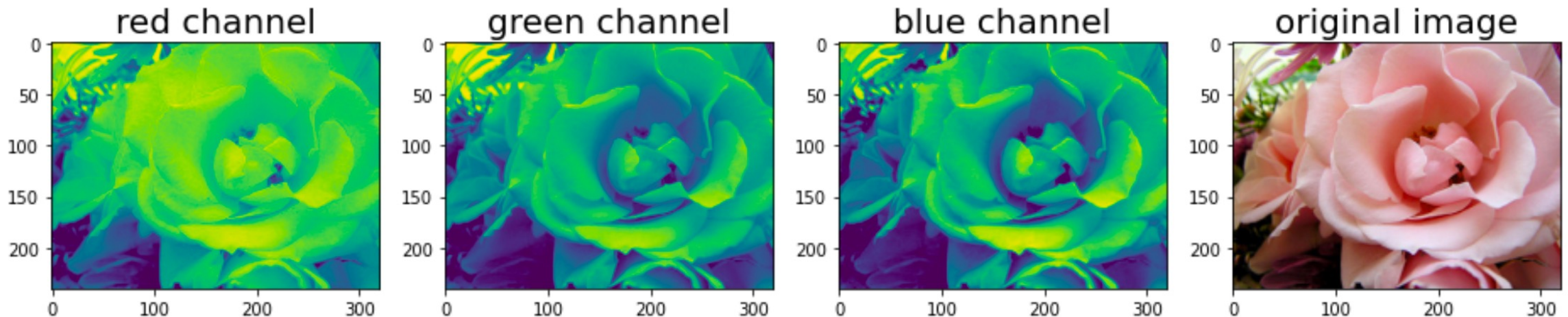
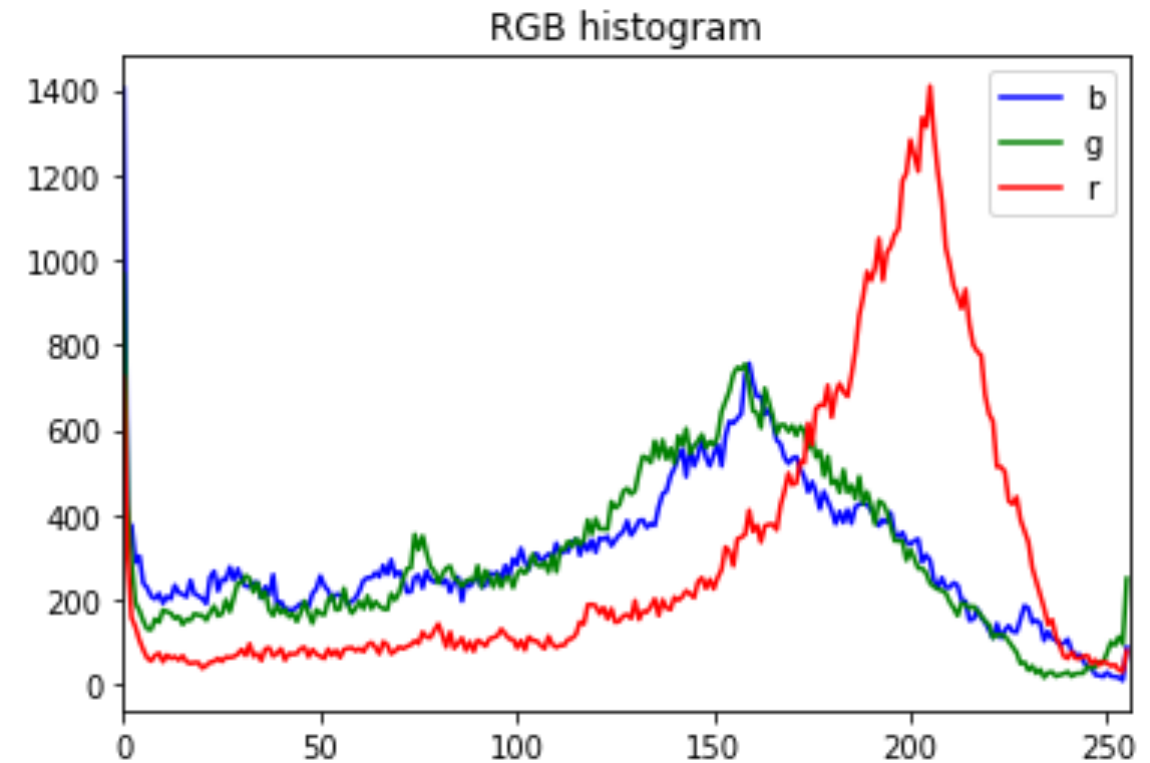
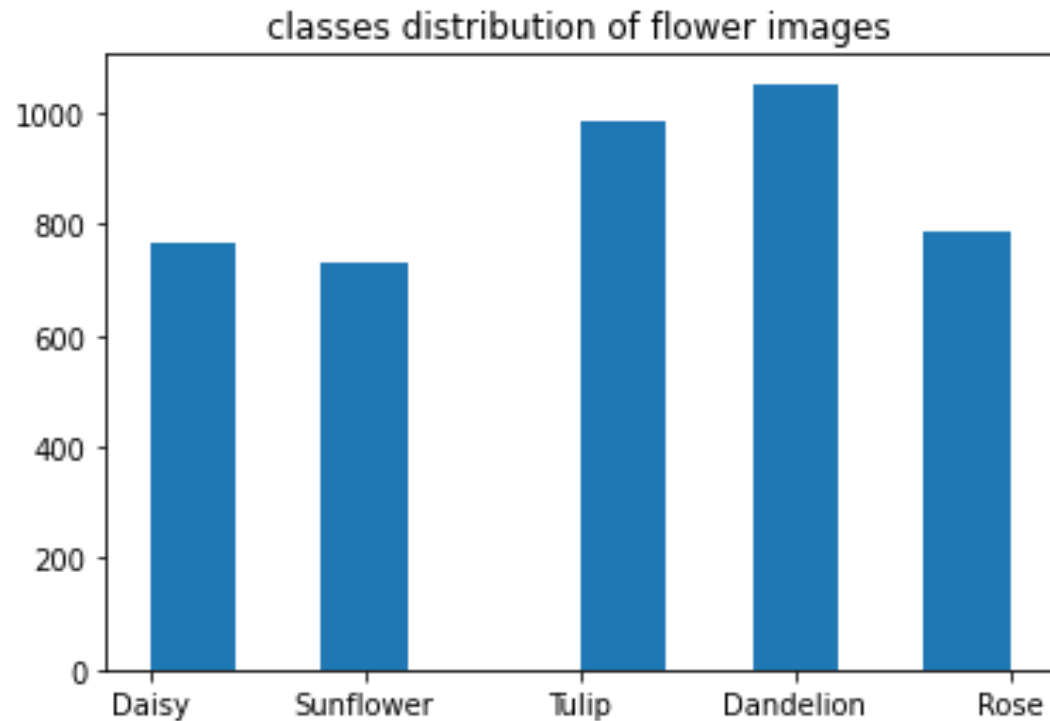


Image feature

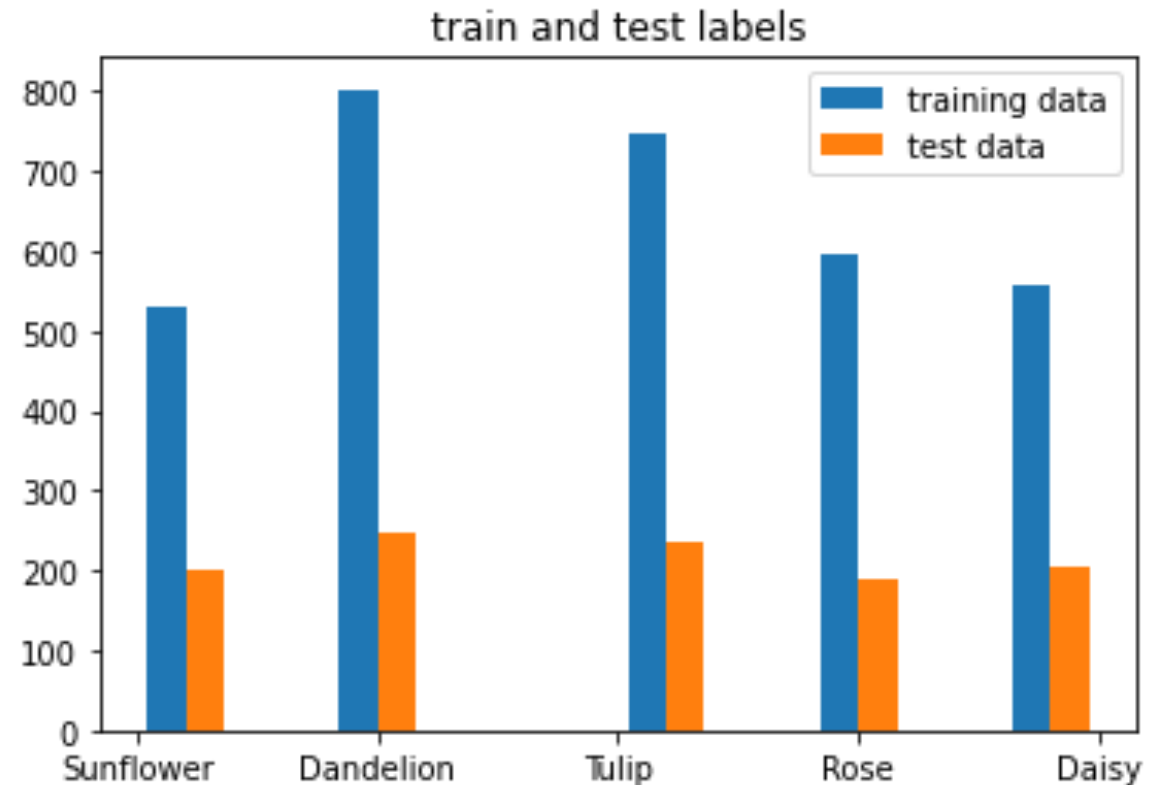
- Function – `extract_color_histogram()`
 - Loading image
 - Resize image as unified size 150×150 with three channels
 - Extract from an image (size of $[150, 150, 3]$) to obtain a color histogram (size of $[6, 6, 6]$)
 - Normalize the elements in histogram (scaling between 0 and 1)
- Loading data – `make_train_data`

Split the data into training, validation and test sets

- Before splitting data



- After data split



Evaluation on MLP classifier

- Performance Metrics
 - Confusion matrix

prediction labels	Daisy	Dandelion	Rose	Sunflower	Tulip
Daisy	91	70	29	4	12
Dandelion	47	149	26	12	15
Rose	20	36	92	2	38
Sunflower	30	40	20	94	17
Tulip	28	54	54	14	86

References

- <https://easychair.org/publications/open/jKTm>
- <https://matplotlib.org/stable/index.html>
- <https://seaborn.pydata.org/introduction.html>
- https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html
- <https://pandas.pydata.org/>
- <https://pypi.org/>
- https://docs.opencv.org/4.x/d5/d26/tutorial_py_knn_understanding.html
- https://keras.io/examples/vision/image_classification_from_scratch/
- <https://cs231n.github.io/convolutional-networks/>
- <https://www.kaggle.com/alxmamaev/flowers-recognition/metadata>

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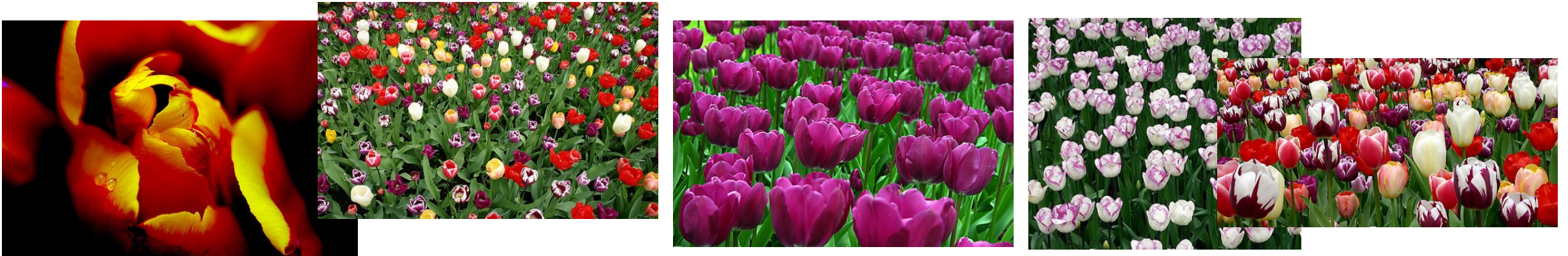


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Machine Learning -- CNN

- Import modules
- Use the images as input to a CNN model
- Divide dataset into training set, validation set, and test set
- Building a CNN model as the classifier
- Evaluation on test set by performance metrics

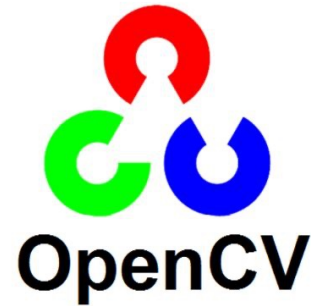
Machine Learning -- CNN

Layer		Feature Map	Size	Kernel Size	Stride	Activation
Input	Image	1	32x32	-	-	-
1	Convolution	6	28x28	5x5	1	tanh
2	Average Pooling	6	14x14	2x2	2	tanh
3	Convolution	16	10x10	5x5	1	tanh
4	Average Pooling	16	5x5	2x2	2	tanh
5	Convolution	120	1x1	5x5	1	tanh
6	FC	-	84	-	-	tanh
Output	FC	-	10	-	-	softmax

<https://www.datasciencecentral.com/lenet-5-a-classic-cnn-architecture/>

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 - Scikit-Learn and **Keras**
- Model performance metrics
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Simple. Flexible. Powerful.

References

- <https://easychair.org/publications/open/jKTm>
- <https://matplotlib.org/stable/index.html>
- <https://seaborn.pydata.org/introduction.html>
- https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html
- <https://pandas.pydata.org/>
- <https://pypi.org/>
- https://docs.opencv.org/4.x/d5/d26/tutorial_py_knn_understanding.html
- https://keras.io/examples/vision/image_classification_from_scratch/
- <https://cs231n.github.io/convolutional-networks/>
- <https://www.kaggle.com/alxmamaev/flowers-recognition/metadata>